Session Progress Report

Delta Robot — Pick & Place Debug & Fix

2026-04-22

Workspace: /home/s4mb4th/delta_ws

1. Session Overview

Goal: Enable gripper-integrated pick-and-place testing on the physical delta robot. On the first trigger, the node crashed immediately with:

```
[ERROR] Sequence aborted: RuntimeError: HOME→above_pick trajectory did not reach target
```

Root-cause analysis revealed **three stacked bugs** in the approach-path calculation. All three were fixed and the full path was re-verified against the inverse-kinematics (IK) solver before any hardware test.

2. Robot Hardware Parameters

Parameter	Symbol	Value	Unit
End-effector platform radius	e	35	mm
Base platform radius	f	157	mm
Forearm (lower arm) length	re	400	mm
Upper arm length	rf	200	mm
HOME position	—	(0, 0, −350)	mm XYZ
Motors	—	CAN IDs 1, 2, 3	RS-00
Pneumatic gripper	—	CAN ID 4	solenoid
Gripper length	gl	120	mm

3. Bug Analysis

Bug 1 — max / min Inversion

In *blind_pick_place.py* → *_run_one()* and *_camera_target_cb()*, the approach-point Z was clamped with **max()** instead of **min()**. Because Z is negative (more negative = deeper), **max** picks the *higher* (less negative) value — the ceiling — rather than staying below it.

```
# BEFORE (wrong) — raises approach to the ceiling:
above_pick = (px, py, max(pz + zoff, config.Z_MAX))

# AFTER (correct) — clamps below the ceiling:
above_pick = (px, py, min(pz + zoff, config.Z_MAX))
```

Concrete example — pick (150, 150, −400), gripper_length=90 mm, offset=80 mm:

Expression	Value	Meaning

pick_z + gl	−400 + 90 = −310	EFF Z above gripper tip
pz + zoff	−310 + 80 = −230	desired approach height
max(−230, −196.875)	−196.875	raises to ceiling (WRONG)
min(−230, −196.875)	−230	stays below ceiling (correct)

Bug 2 — config.Z_MAX was wrong by 127 mm

The workspace ceiling constant in *delta_common/config.py* was set to –**196.875 mm** — a theoretically derived value that does not match the actual IK solver. The correct formula for the ceiling at the robot centre is:

```
Z_MAX = -sqrt(re^2 - (rf + (f-e)*tan30/2)^2)
      = -sqrt(400^2 - (200 + 35.22)^2)
      = -sqrt(160000 - 55328)
      ≈ -323.5 mm
```

Config key	Old value	Correct value	Error
Z_MAX (ceiling)	−196.875 mm	−323.0 mm	127 mm too high

Any Z value above –323 fails IK. The old value of –196.875 caused check_workspace() to accept 127 mm of unreachable space above the robot.

Bug 3 — 'Approach from above' is geometrically impossible

HOME is only **26 mm below the actual IK ceiling** (–350 vs –324). The approach offset (gripper_length 90 mm + z_offset 80 mm = 170 mm) always pushes the approach point 144 mm *above* the ceiling — permanently unreachable.

Additionally, the workspace is **non-convex near the ceiling**: a diagonal path from HOME(0,0,–350) toward (150,150,–400) passes through a dead zone around (75–120, 75–120, –375 to –395) where IK has no solution. A direct trajectory stalls mid-path.

Reachability — key points along old vs new path

Point	Coordinates (mm)	Context	IK	check_ws
HOME	(0, 0, −350)	start / end	OK	True
above_pick OLD	(150, 150, −196.875)	old approach (Bug 1+2)	FAIL	False
above_pick fixed	(150, 150, −230)	after min-fix (still bad)	FAIL	False
pick	(150, 150, −400)	actual pick	OK	True
home_t (transit)	(0, 0, −400)	new path waypoint	OK	True
pick_t (transit)	(150, 150, −400)	new path waypoint	OK	True
place_t (transit)	(0, −80, −400)	new path waypoint	OK	True
place (new default)	(0, −80, −380)	new drop position	OK	True

4. Workspace Shape

The delta robot workspace is **not rectangular**. config.X_LIMIT and Y_LIMIT (151.563 mm) are a bounding-box over-approximation. The actual cross-section is a three-lobed shape aligned with the three arms. The workspace also thins severely near the ceiling.

Direction	Arm relation	Max R at Z=–380	Max R at Z=–400
+X (0°)	between arms	161 mm	≥200 mm
+Y (90°)	opposite arm 1	200 mm	≥200 mm
–Y (270°)	arm 1 direction	110 mm	≥200 mm
+X+Y (45°)	diagonal	~110 mm	≥200 mm
Centre (0,0)	—	ceiling at Z = –324	—
At (0, 50)	+Y	ceiling at Z = –340	—
At (0, –80)	–Y	ceiling at Z = –370	—

5. Fix Summary

File	Change	Lines affected
delta_common/config.py	Z_MAX: –196.875 → –323.0 (correct IK ceiling)	Line 162
delta_common/fk_ik.py	check_workspace() now also calls delta_calcInverse() to reject reachable-box corners that fail IK	Lines 125–138
delta_main_app/blind_pick_place.py	max() → min() in _camera_target_cb() approach-Z clamp (2 occurrences)	Lines 284, 292
delta_main_app/blind_pick_place.py	run_one() rewritten: replaced 'approach from above' with safe transit strategy (HOME → centre column at transit_z → lateral traverse → pick/place)	Lines 365–445
delta_main_app/blind_pick_place.py	Default drop: (0, –80, –300) → (80, 0, –380) [old drop was unreachable]	Lines 110–112

6. New Path Algorithm (Transit Strategy)

The new *_run_one()* always routes through the **centre column (HOME_X=0, HOME_Y=0)** at a transit Z that is deep enough for the full workspace to be open. A hard floor of –400 mm guarantees ≥200 mm radius in every direction.

```
SAFE_TRANSIT_Z = -400.0
transit_z = min(pick_xyz[2], place_xyz[2], SAFE_TRANSIT_Z)

home_t  = (0,        0,        transit_z)
pick_t  = (pick_x,  pick_y,  transit_z)
place_t = (place_x, place_y, transit_z)

# PICK
HOME → home_t       # descend at centre
home_t → pick_t     # lateral traverse at safe depth
pick_t → pick       # final descent (zero if pick_z == transit_z)
GRASP

# PLACE
pick → pick_t       # rise
pick_t → home_t     # back to centre (avoids non-convex diagonal gaps)
home_t → place_t    # traverse to place
place_t → place     # final descent
DROP

# RESET
place → place_t → home_t → HOME
```

7. Verified Path — pick(150,150,–400) → place(0,–80,–380)

All 11 waypoints verified reachable (IK st=0) in simulation before hardware test:

#	Label	Coordinates (mm)	Thetas θ■, θ■, θ■ (°)	IK
1	HOME	(0, 0, –350)	(7.3, 7.3, 7.3)	OK
2	home_t	(0, 0, –400)	(19.9, 19.9, 19.9)	OK
3	pick_t	(150, 150, –400)	(58.1, 1.4, 46.0)	OK
4	pick	(150, 150, –400)	(58.1, 1.4, 46.0)	OK
5	pick_t	(150, 150, –400)	(58.1, 1.4, 46.0)	OK
6	home_t	(0, 0, –400)	(19.9, 19.9, 19.9)	OK
7	place_t	(0, –80, –400)	(8.7, 28.7, 28.7)	OK
8	place	(0, –80, –380)	(3.2, 24.2, 24.2)	OK
9	place_t	(0, –80, –400)	(8.7, 28.7, 28.7)	OK
10	home_t	(0, 0, –400)	(19.9, 19.9, 19.9)	OK
11	HOME	(0, 0, –350)	(7.3, 7.3, 7.3)	OK

8. How to Run the Test

Terminal 1 — launch hardware nodes

```
cd ~/delta_ws && source install/setup.bash
ros2 launch delta_main_app blind_pick_place.launch.py
```

Terminal 2 — interactive UI

```
ros2 run delta_main_app pick_place_ui
```

Or trigger directly (default positions)

```
ros2 service call /delta/trigger_pick std_srvs/srv/Empty
```

Custom pick point via topic

```
ros2 topic pub --once /delta/blind_target \
  geometry_msgs/msg/PointStamped \
  "{point: {x: 150.0, y: 150.0, z: -400.0}}"
```

Monitor FSM state

```
ros2 topic echo /delta/robot_state
```

9. Files Changed (git diff summary)

File	Type of change
src/delta_common/delta_common/config.py	Z_MAX constant fixed
src/delta_common/delta_common/fk_ik.py	check_workspace now includes IK feasibility test
src/delta_main_app/delta_main_app/blind_pick_place.py	max-min fix; _run_one rewritten; default positions updated